
Reward-Shaping-Free Deep Market Making via Avellaneda–Stoikov Policy Regularization

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Abstract

Deep reinforcement learning (RL) market makers are often trained with hand-engineered rewards that mix dampened profit-and-loss, transaction advantage, and inventory penalties. We revisit the continuous-action market maker of Guo et al. [7] (Attn-LOB encoder with C-PPO) and ask whether the same behavioural structure can instead be supplied by an interpretable action-space prior. We propose **AS-regularized PPO**: the base reward is pure profit, and a soft penalty keeps the quote action close to a per-stock Avellaneda–Stoikov (AS) teacher fitted on training data. On a paper-faithful synthetic replication with per-stock models trained for 400k steps over three seeds, our selected low-risk soft-AS variant improves mean episode profit over reward-shaped C-PPO by 14.28 (paired 95% CI [8.25, 20.31], $p = 6 \times 10^{-4}$, 9/9 stock-seed wins) and over a profit-only PPO control by 7.93 ($p = 0.011$). Profit-only PPO already improves on C-PPO, but the AS prior adds a further statistically significant gain. In the matched batch, soft regularization also outperforms hard AS action constraints; in exploratory runs, it outperformed BC warm-start. The benchmark is synthetic, statistics rest on nine paired units, and latency/divergence diagnostics remain the most important next checks.

1 Introduction

Market making (MM) is the problem of continuously quoting bid and ask limit orders to capture the spread while controlling inventory risk [17, 12]. The limit order book (LOB) is a high-dimensional, low signal-to-noise data stream, and modelling it well is the central difficulty in learning a market-making policy [5]. Classical models such as Avellaneda–Stoikov (AS) [1] solve a stochastic-control formulation in closed form but require strong assumptions about the price and order-arrival processes. Deep RL removes those assumptions by learning directly from data, and the recent Attn-LOB / C-PPO agent of Guo et al. [7] is a strong representative: a convolution-plus-attention encoder reads a 50×40 LOB window, and a PPO policy emits AS-like quotes through a continuous action space.

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A recurring feature of these systems is the *reward*. Guo et al. [7], following Spooner et al. [16], do not optimize profit directly. They optimize a hybrid reward

$$R_t = \underbrace{\Delta \text{PnL}_t - \max(0, \eta \Delta \text{PnL}_t)}_{\text{dampened PnL (DP)}} + \underbrace{X_{t,v} (p_{m,t} - X_{t,p})}_{\text{trading PnL (TP)}} - \underbrace{\zeta \text{Inv}_t^2}_{\text{inventory penalty (IP)}}, \quad (1)$$

where each term is a hand-designed proxy for a behaviour the designer wants: dampen speculative holding, reward favourable fills, discourage large inventory. This works, but it has well-known costs. The terms are not potential-based, so they change the optimal policy rather than merely accelerating learning [10]; the coefficients η, ζ and the implicit TP weight interact; and the recipe must be re-tuned per market. The learned behaviour is, in a real sense, an artifact of the penalty design as much as of the data.

This paper asks a simple question: *can we recover the good behaviour of a reward-shaped market maker without shaping the reward?* Our answer separates two things that Eq. (1) entangles. The *economic objective* is profit. The *behavioural prior* — quote symmetrically around a risk-adjusted reservation price with a sensible spread — is exactly what the AS model encodes. We therefore keep profit as the only reward and add a soft regularizer that penalizes deviation, in action space, from a per-stock AS teacher fitted on the training data:

$$R_t^{\text{train}} = \Delta \text{PnL}_t - \lambda \|a_t^\pi - a_t^{\text{AS}}\|_2^2. \quad (2)$$

The single coefficient λ trades profit against adherence to a theoretically grounded prior. The prior is also interpretable: diagnostic evaluation can log the teacher action, the policy action, and their divergence at every step.

Contributions. (i) We cast AS as a *policy-space regularizer* for continuous-action PPO market making, with a base reward of pure profit (Section 4). (ii) We give a per-stock empirical calibration pipeline that fits the AS teacher’s (γ, κ) from training-window statistics (Section 4, Appendix B). (iii) On a controlled synthetic benchmark that matches the original paper’s stock/day/episode/action design, with matched 400k-step per-stock training over three seeds, we show that the selected soft-AS policy significantly improves over both reward-shaped C-PPO and a profit-only PPO control under paired statistics (Section 6). (iv) We compare the proposed soft prior with hard AS constraints in the matched batch and with BC warm-start in exploratory runs (Sections 6–7). (v) We state the limitations plainly and identify the latency and teacher-divergence diagnostics needed before any live-market claim (Section 8).

We emphasize the scope of our claim up front. We do **not** claim that AS regularization universally dominates reward shaping. We claim that, on a paper-faithful synthetic setup, soft AS regularization yields a reward-shaping-free PPO market maker that is significantly stronger than reward-shaped C-PPO and than profit-only PPO, while retaining an explicit link to classical theory.

2 Background and related work

Classical market making. Ho and Stoll [8] and Avellaneda and Stoikov [1] formulate market making as stochastic optimal control. For inventory q , mid-price s , volatility σ , risk aversion γ , liquidity κ , and horizon T , the AS reservation price and optimal total spread are

$$r(s, q, t) = s - q\gamma\sigma^2(T - t), \quad \delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{\kappa}\right). \quad (3)$$

These closed forms are interpretable and remain a standard benchmark, but they depend on a parametric model of order flow [6]. Gašperov and Kostanjčar [4] instead let an RL agent tweak the AS risk-aversion parameter; our regularizer is complementary, treating the full AS quote as a target in action space rather than learning a scalar inside AS.

Deep learning on LOBs and DRL market making. CNN/RNN/attention encoders for LOB data were established by DeepLOB and related models [19, 18, 11]. Earlier RL market makers used handcrafted features and discrete actions [9, 16, 20, 14, 3]; Guo et al. [7] combined an attention LOB encoder with a continuous, AS-inspired action space and the hybrid reward of Eq. (1). We retain their encoder, action space, and simulator, and change only the reward/regularization.

Reward shaping versus regularization. Potential-based reward shaping is policy-invariant [10]; the DP/TP/IP terms in Eq. (1) are not potential-based and therefore alter the objective. An alternative to encoding behaviour in the reward is to encode it as a prior over policies. This connects to imitation and offline-regularized RL, where a policy is kept close to a reference via behavioural cloning or a KL/action penalty [15, 13, 2]. Our method is the market-making instance of this idea: the reference is a *model-based* AS teacher, not a logged dataset, and the penalty lives in the continuous quote-action space.

3 The replicated agent

We summarize the components we inherit from Guo et al. [7]; full details are in the original paper and our reproduction notes in Appendix A.

State and encoder. The agent observes a rolling window of $T = 50$ LOB snapshots, each a 40-dimensional vector of the top-10 ask/bid prices and volumes. The Attn-LOB encoder applies convolutional filters, an Inception module for multi-scale temporal structure, and a multi-head self-attention layer. Dynamic agent state (inventory, remaining time, cash/value) is appended. The encoder is pre-trained on a mid-price direction task and reused as the PPO feature extractor.

Continuous action space. The policy emits two actions $A_1, A_2 \in [0, 1]$. A_1 sets a reservation-price bias and A_2 the quoted spread:

$$\delta = A_1 \cdot \text{max_bias}, \quad p_r = p_m - \text{sign}(\text{Inv}) \delta, \quad \text{spread} = A_2 \cdot \text{max_spread}, \quad p_{a,b} = p_r \pm \frac{\text{spread}}{2}. \quad (4)$$

This parameterization is deliberately AS-like, which is what makes an AS *teacher* a natural prior in the same coordinates. We keep $\text{max_bias} = 0.05$, $\text{max_spread} = 0.1$, $\text{minimum_trade_unit} = 100$, and $\text{max_inventory} = \omega \cdot \text{minimum_trade_unit}$ with $\omega = 10$.

Reward (baseline). The reward-shaped baseline, which we call C-PPO, uses the hybrid reward of Eq. (1) with $\eta = 0.5$, $\zeta = 0.01$, trained with PPO. This is our primary point of comparison.

Simulator. A replay simulator fills the agent’s quotes only when the recorded stream trades through the quoted price. There is no market impact and zero transaction cost; the agent may hold positive or negative inventory and liquidates at end of episode (2000 events). This mirrors the small-order assumption of the original work.

4 Method: Avellaneda–Stoikov policy regularization

4.1 The AS teacher in action space

At each step we convert the AS closed form of Eq. (3) into a target in the policy’s normalized action coordinates of Eq. (4). Given the current inventory q , a running volatility estimate $\hat{\sigma}$, remaining time $T - t$, and fitted parameters (γ, κ) , the AS reservation-price bias is $\delta^{\text{AS}} = |q| \gamma \hat{\sigma}^2 (T - t)$ and the AS half-spread follows Eq. (3). Projecting into $[0, 1]^2$,

$$a_t^{\text{AS}} = \left(\text{clip}\left(\frac{\delta^{\text{AS}}}{\text{max_bias}}, 0, 1\right), \text{clip}\left(\frac{\delta^a + \delta^b}{\text{max_spread}}, 0, 1\right) \right). \quad (5)$$

The teacher is a fixed function of state once (γ, κ) are calibrated; it adds no learnable parameters and is cheap to evaluate (Appendix B).

4.2 Soft regularization, hard constraints, and controls

We study a family of training signals that share the *pure-profit* base reward ΔPnL_t and differ only in how they use the teacher.

Soft-AS (proposed). Add the action-space penalty of Eq. (2). The policy is free to deviate from a^{AS} wherever deviating is profitable, but pays a quadratic cost for doing so. A single coefficient λ controls the strength of the prior.

Figure 1. Paper architecture with AS-regularized extension overlay

The original policy/action/simulator path is retained; the proposed extension replaces ad hoc reward shaping with a train-data-fitted AS teacher penalty.

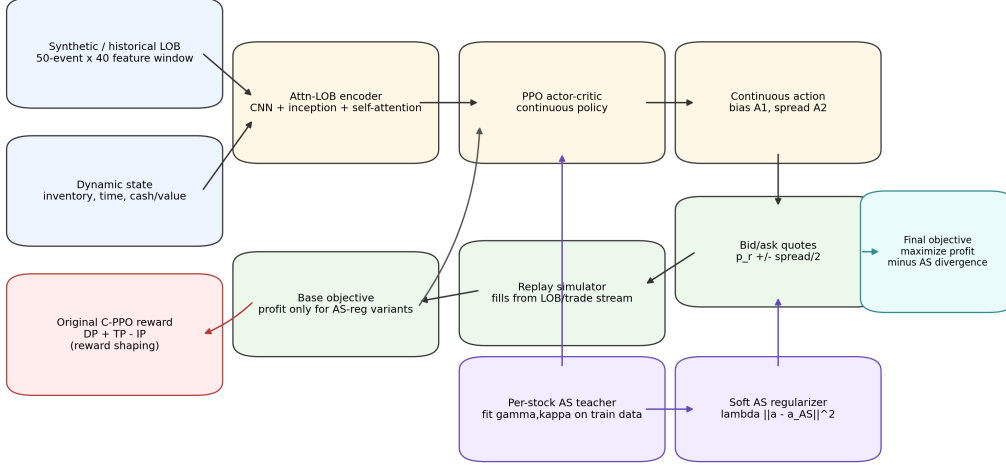


Figure 1: The proposed agent retains the original Attn-LOB encoder, PPO policy, continuous action space, and replay simulator (yellow/green path). The change is the training signal: the base reward becomes pure profit, and a per-stock AS teacher fitted on training data supplies a soft penalty on the policy’s quote action (purple). The original hybrid reward (red) is removed.

Hard-AS (ablation). Constrain the policy action to a fixed window of half-width w around the teacher, $a_t^\pi \in [a_t^{\text{AS}} - w, a_t^{\text{AS}} + w]$, by clipping. This enforces AS-like quoting but removes adaptivity.

Profit-PPO (control). Drop the teacher entirely ($\lambda = 0$). This isolates the effect of removing the original reward shaping from the effect of adding the AS prior, and is essential for attributing any improvement correctly.

BC warm start (control). Behaviour-clone the policy on AS demonstrations before PPO fine-tuning. The cloning phase is supervised imitation only, with no environment reward. The subsequent PPO phase uses pure profit with no standing AS penalty. This tests whether an AS *initialization* suffices without a persistent prior.

4.3 Per-stock teacher calibration

We fit one teacher per stock from its training window. We estimate κ from the mean level-1 half-spread and γ from episode-level volatility; the resulting per-stock values are $\kappa \in \{125.2, 23.9, 83.8\}$ for the three instruments, reflecting their differing liquidity, with a base $\gamma = 0.05$ (Appendix B). We additionally consider a *low-risk* teacher that halves γ to 0.025 and a *high-risk* teacher that doubles it, plus alternative intensity calibrations (spread- κ , fill- κ). As Section 6 shows, teacher calibration changes behaviour more than small changes in λ do.

5 Experimental setup

Synthetic benchmark. The original Shenzhen order-flow data are proprietary. We therefore build a synthetic event-level LOB generator that preserves the *experimental shape* of the paper: the same three stock identities (a low-price banking-like name 000001, a high-price liquid name 000858, a mid-price technology-like name 002415), a fair-value process with bid–ask bounce and stable/trending/volatile regimes, top-10 levels, the paper’s stable intraday windows, 10 train days and 11 test days, and 2000-event episodes. Details and validity caveats are in Appendix A. We are explicit that this controlled benchmark supports method-comparison claims, not live-market performance claims.

Matched protocol. Following the original paper, all learned models are *per-stock*, not a single cross-stock policy. The full extension batch trains 12 method specifications $\times$ 3 stocks $\times$ 3 seeds (plus one direct-AS baseline per stock) for 400,000 PPO steps each, one GPU per task; the paper reports the selected proposed setting and the controls needed to interpret it. Every method sees identical data, episodes, action space, and trade granularity; only the training signal differs. The implementation keeps the paper replication code and the extension code separated: AS-guided methods live under `src/mlfcs_gapa/extensions/`.

Metrics. We report mean episode profit (PnL), spread-normalized ND-PnL, Sharpe ratio, mean absolute inventory, and mean quoted spread. For an episode with final marked-to-market value V_T and initial value V_0 , $\text{PnL} = V_T - V_0$ and $\text{ND-PnL} = \text{PnL}/(\bar{s} + 10^{-7})$, where $\bar{s}$ is the mean quoted spread. Sharpe is annualization-free and computed on event-to-event marked-value increments ΔV_t as $\sqrt{n} \Delta \bar{V} / \text{sd}(\Delta V)$. Because summed PnL is dominated by the high-price stock, we report per-stock results and aggregate *paired* statistics, never a single averaged score alone.

Paired statistics. The natural paired unit is (stock $\times$ seed), giving 9 units per PPO method. For each proposed method we report the paired mean PnL difference against C-PPO and against PROFIT-PPO, a paired- t 95% confidence interval, the paired p -value, and the win rate (fraction of units that beat the baseline). Nine units is small; we treat the statistics as strong evidence for a direction, not as a final verdict (Section 8).

6 Results

Table 1: Aggregate test performance on the matched 400k benchmark. Learned methods are averaged over 9 stock–seed units; direct AS is averaged over one deterministic run per stock. PnL is reported with its standard error.

Method	PnL ($\pm$ SE)	ND-PnL	Sharpe	Abs. inv.	Quoted spread
SOFT-AS _{low} ^{0.10} (proposed)	73.96 $\pm$ 28.4	1220.6	2.50	56.5	0.0465
PROFIT-PPO (control)	66.03 $\pm$ 29.7	1029.6	2.37	27.2	0.0537
Hard-AS $w=0.30$	61.66 $\pm$ 26.3	895.7	2.41	26.0	0.0557
C-PPO (reward-shaped baseline)	59.68 $\pm$ 27.6	806.8	2.13	19.7	0.0624
AS direct (fitted)	21.80 $\pm$ 7.0	422.2	0.76	102.7	0.0567

6.1 Soft AS regularization beats reward shaping

Table 1 gives the aggregate picture for the selected methods. The proposed SOFT-AS_{low}^{0.10} attains the highest Sharpe among learned policies (2.50 vs. 2.13 for C-PPO) and a 24% PnL improvement over the reward-shaped baseline. It also improves on the profit-only control, which is the central attribution test: the gain is not only from removing the old hybrid reward.

The paired comparison against C-PPO (Table 2) makes the result statistically concrete. SOFT-AS_{low}^{0.10} improves mean PnL by 14.28 with a paired 95% CI of [8.25, 20.31], $p = 6.0 \times 10^{-4}$, and a 100% win rate: every one of the nine stock–seed units beats the reward-shaped baseline. Against PROFIT-PPO, the same method improves mean PnL by 7.93 (95% CI [2.43, 13.43], $p = 0.011$).

Table 2: Paired PnL differences (stock–seed units, $n=9$) against reward-shaped C-PPO and the profit-only PPO control.

Method	vs. reward-shaped C-PPO				vs. profit-only PPO			
	Δ PnL	95% CI	p	win	Δ PnL	95% CI	p	win
SOFT-AS _{low} ^{0.10} (proposed)	14.28	[8.25, 20.31]	0.0006	1.00	7.93	[2.43, 13.43]	0.011	0.89
PROFIT-PPO	6.35	[−1.13, 13.84]	0.086	0.78	—	—	—	—
Hard-AS $w=0.30$	1.98	[−2.47, 6.43]	0.335	0.78	−4.37	[−13.71, 4.96]	0.311	0.56

6.2 The improvement is not just “removing bad shaping”

The single most important control is PROFIT-PPO: PPO on pure profit, no teacher. It is strong — mean PnL 66.03, above C-PPO’s 59.68 — which already shows that part of C-PPO’s deficit comes from the shaping itself (the paired difference PROFIT-PPO vs. C-PPO is +6.35, though only marginally significant, $p = 0.086$). If soft-AS merely matched PROFIT-PPO, the paper would reduce to “the original reward was suboptimal.” Instead, the right-hand block of Table 2 shows that the AS prior contributes signal *beyond* cleaning up the objective. Note also that hard constraints and C-PPO both *lose* to PROFIT-PPO, underlining that the gain comes from a soft prior, not from any use of AS.

6.3 Per-stock behaviour and the teacher-policy relationship

Aggregate PnL is dominated by the high-price stock 000858, so Appendix E reports per-stock means. The ranking is not driven by only one name: on 000858, $\text{SOFT-AS}_{\text{low}}^{0.10}$ (186.4) beats both PROFIT-PPO (183.5) and C-PPO (169.6); on 002415, $\text{SOFT-AS}_{\text{low}}^{0.10}$ (25.8) roughly doubles PROFIT-PPO (12.1) and triples C-PPO (8.7); on 000001, $\text{SOFT-AS}_{\text{low}}^{0.10}$ (9.7) leads the learned baselines. Direct empirical AS is weak in aggregate (PnL 21.8, Sharpe 0.76) and carries the most inventory (102.7), though it performs competitively on 000001 and 002415. This argues against a trivial “just copy AS” explanation, but a dedicated teacher-divergence analysis is still needed to show exactly where and why the learned policy departs from the teacher.

6.4 Participation and risk diagnostics

Soft-AS earns more by participating more. Relative to C-PPO, it quotes tighter (0.0465 vs. 0.0624 in Table 1), fills more often, and trades more volume (Table 3). The cost is higher inventory and larger within-episode drawdowns. This is a market-making-like tradeoff rather than a free improvement: the proposed policy is more active and more profitable, but the risk budget must be reported alongside PnL.

Table 3: Participation and risk diagnostics from the same matched 400k evaluations. Drawdown is the mean within-episode decline from the running maximum of marked-to-market value. Liquidation PnL is the mean value change on forced end-of-episode close.

Method	Fills/episode	Abs. volume	Max abs. inv.	Drawdown	Liquidation PnL	Zero-fill
$\text{SOFT-AS}_{\text{low}}^{0.10}$	516.8	51,800	444.4	97.8	−0.75	0.00
PROFIT-PPO	460.3	46,067	255.6	76.5	0.00	0.00
Hard-AS $w=0.30$	415.7	41,600	188.9	63.5	0.00	0.00
C-PPO	335.8	33,600	177.8	58.1	0.00	0.00
AS direct (fitted)	245.8	26,100	766.7	105.6	−4.00	0.00

6.5 Calibration matters more than the exact penalty weight

The main text uses a single proposed setting, $\text{SOFT-AS}_{\text{low}}^{0.10}$, selected for its risk-adjusted profile. Appendix E summarizes the broader soft-penalty sweep. Across the base-teacher sweep, PnL remained above the C-PPO line for $\lambda \in \{0.03, 0.05, 0.10, 0.125, 0.20\}$, with the broad best region around 0.05–0.125. Inventory does not increase monotonically over the whole sweep, but soft-AS variants generally carry more inventory than C-PPO because they quote tighter and participate more. Changing the teacher has a first-order effect: at fixed $\lambda = 0.10$, halving γ (low-risk) raises Sharpe to 2.50 and lowers inventory to 56.5, whereas doubling γ pushes inventory to 81.4 and Sharpe down to 2.36. The practical message is that the penalty weight is forgiving within a reasonable range, while teacher specification deserves care.

7 Discussion

Why does a soft prior beat both shaping and a hard constraint? The three training signals occupy different points on a spectrum of how much structure they impose. Reward shaping (Eq. 1)

imposes structure *indirectly* through several non-potential penalty terms; the learned policy is conservative and wide-spread here, leaving PnL on the table. A hard AS constraint imposes structure *rigidly* in action space; it suppresses the profitable deviations that the data clearly support, so it barely beats C-PPO and loses to PROFIT-PPO. Soft AS imposes structure *softly*: it nudges the policy toward a theoretically sensible quoting region while leaving room to depart where departing pays. The empirical ordering — soft > profit-only > hard $\approx$ shaped — is consistent with this reading.

Interpretability. Because teacher and policy actions can be logged at every step, the method is legible in a way reward shaping is not: one can ask, per state, how far the policy strays from AS and whether straying paid. Attention maps and a decision trace (Appendix D, Figs. 3–4) show the encoder concentrating on the most recent events and the policy opening/closing around a drifting fair value while bounding inventory. We treat these as diagnostics, not proof of mechanism; a full divergence analysis is planned.

Limitations. We state these plainly. (1) **Synthetic data.** The benchmark is controlled and proprietary-data-free, which is ideal for method development but limits external validity; no claim about live or proprietary-LOB performance is warranted. (2) **Sample size.** Paired statistics rest on nine stock-seed units. The $\text{SOFT-AS}_{\text{low}}^{0.10}$ result is strong ($p = 6 \times 10^{-4}$, full win rate), but more seeds would tighten it. (3) **Latency.** We measured the proposed method only at latency = 1; a full latency sweep is not yet run (Fig. 2 overlays the single measured point on the original sweep). (4) **Inventory risk.** Soft-AS carries more inventory than C-PPO. This is consistent with more active liquidity provision, but inventory, drawdown, and liquidation cost must be reported alongside PnL, not after it. (5) **Attention.** Visualizations are suggestive, not causal.

8 How to strengthen this work

The present results justify the *direction*; the following experiments would make the claim publication-grade, in rough priority order.

1. **Full latency sweep** for $\text{SOFT-AS}_{\text{low}}^{0.10}$, C-PPO, and PROFIT-PPO at identical checkpoints across latencies $\{1, 5, 10, 20, 50\}$ events, replacing the single-point Fig. 2 with a true robustness curve.
2. **Evaluation-only teacher diagnostics:** rerun deterministic evaluation for saved checkpoints while logging a_t^π , a_t^{AS} , and $\|a_t^\pi - a_t^{\text{AS}}\|$ by stock, regime, inventory bucket, and fill outcome.
3. **More seeds** for the selected methods, starting with two additional seeds for $\text{SOFT-AS}_{\text{low}}^{0.10}$, C-PPO, and PROFIT-PPO, to shrink the paired intervals.
4. **Risk diagnostics by episode:** the main table now includes fills, drawdown, liquidation PnL, and max inventory; the next version should add per-episode distributions and tail quantiles.
5. **Proposed-method ablations:** no teacher (=PROFIT-PPO), frozen vs. trainable encoder, no encoder pre-training, and alternative teacher-divergence metrics.
6. **A semi-real validation** on any non-proprietary LOB source (e.g. a public crypto or equities feed), which is the only way to move beyond synthetic claims.
7. **An adaptive or state-dependent λ** (annealed, or larger in volatile regimes), turning the single knob into a curriculum.

9 Conclusion

We revisited a reward-shaped deep market maker and showed that its behaviour can be recovered, and improved, by moving structure out of the reward and into a soft, model-based prior. AS-regularized PPO keeps profit as the only reward and pulls the quote policy toward a per-stock Avellaneda–Stoikov teacher fitted on training data. On a paper-faithful synthetic benchmark with matched 400k per-stock training, the selected low-risk soft-AS variant significantly beats both reward-shaped C-PPO and a profit-only PPO control, and it outperforms hard AS constraints in the matched batch. The method trades reward engineering for a single regularization weight and a calibrated teacher; our results suggest that, at least in this controlled setting, that trade is worth making. The honest next step is to stress-test it under latency, teacher-divergence diagnostics, more seeds, an explicit risk budget, and non-synthetic data.

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A Synthetic benchmark and reproduction notes

The original event-by-event Shenzhen Stock Exchange data are proprietary, so we generate synthetic LOB days that preserve the paper’s experimental structure. The generator produces a fair-value process with transient bid–ask bounce, intraday stable/trending/volatile regimes, market-order flow, top-10 price/volume levels, and trade extrema. Each synthetic day contains 10,000 raw events before stable-window filtering; the downstream filter keeps only the paper’s stable trading windows (10:00–11:30, 13:00–14:30). We then split 10 train and 11 test days. The three synthetic instruments use base prices 16.45 (000001, banking-like), 130.00 (000858, high-price/liquid), and 35.00 (002415, technology-like). The replay simulator fills the agent’s quotes only when the recorded stream trades through the quote, with no market impact, zero transaction cost, signed inventory, and end-of-episode liquidation.

The Attn-LOB encoder is reused unchanged as the PPO feature extractor. On the synthetic mid-price-direction pre-training task, the exact F1 values of the original (proprietary-data) Table I are not reproduced; the synthetic classification task is different and weaker. We therefore treat the encoder as a reproduced *architecture* component rather than re-stating the original pre-training numbers. This synthetic pipeline is appropriate for controlled method comparison; it does not license live-market claims.

B AS teacher derivation and calibration

The teacher maps state to a target action via Eq. (5), using the AS reservation price and optimal spread of Eq. (3). Parameters are fit per stock from the training window: κ from the mean level-1 half-spread (an order-arrival intensity proxy) and γ from episode-level realized volatility, giving $\kappa = 125.2, 23.9, 83.8$ for 000001, 000858, 002415 respectively and base $\gamma = 0.05$. The low-risk teacher uses $\gamma = 0.025$, the high-risk teacher $\gamma = 0.10$. Spread- κ and fill- κ variants explore alternative intensity calibrations. The teacher adds no learnable parameters and is evaluated in closed form, so its training-time cost is negligible relative to the PPO update; direct AS evaluation is far cheaper than a PPO forward pass.

C Latency

Figure 2 overlays the proposed method’s measured latency= ∞ point on the original C-PPO/D-DQN/AS latency sweep from our replication. We deliberately do *not* interpolate or invent proposed-method values at higher latencies; a full sweep is listed as a required follow-up (Section 8).

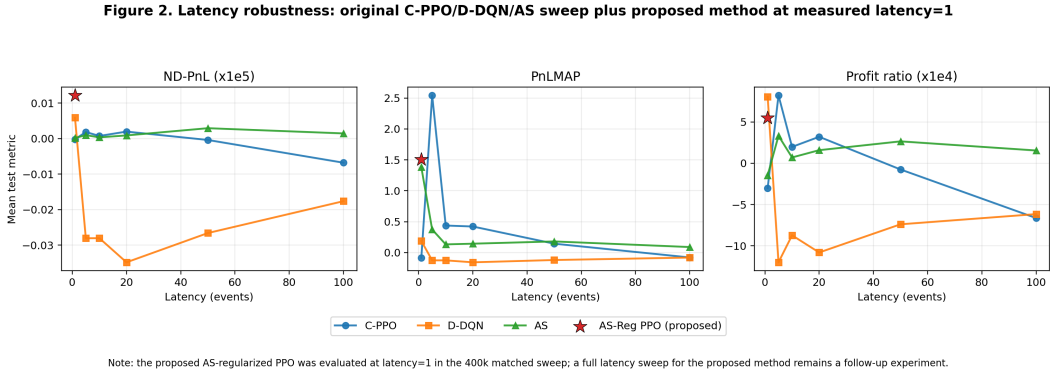


Figure 2: Latency robustness of the replicated baselines (C-PPO, D-DQN, AS) with the proposed AS-regularized PPO shown at its single measured latency= ∞ operating point (star). A full proposed-method latency sweep remains future work.

D Interpretability diagnostics

Figure 3 visualizes the Attn-LOB self-attention weights for the proposed SOFT-AS_{low}^{0.10} policy: in stable markets attention concentrates on the most recent events, while in rapidly changing markets it also attends to earlier structure. Figure 4 traces one test episode (stock 000858, seed 2): the policy opens and closes around a drifting fair value, skewing quotes to bound inventory, with cumulative value rising over the episode. These are diagnostics of *what* the encoder and policy do, not causal explanations of *why* PnL improves.

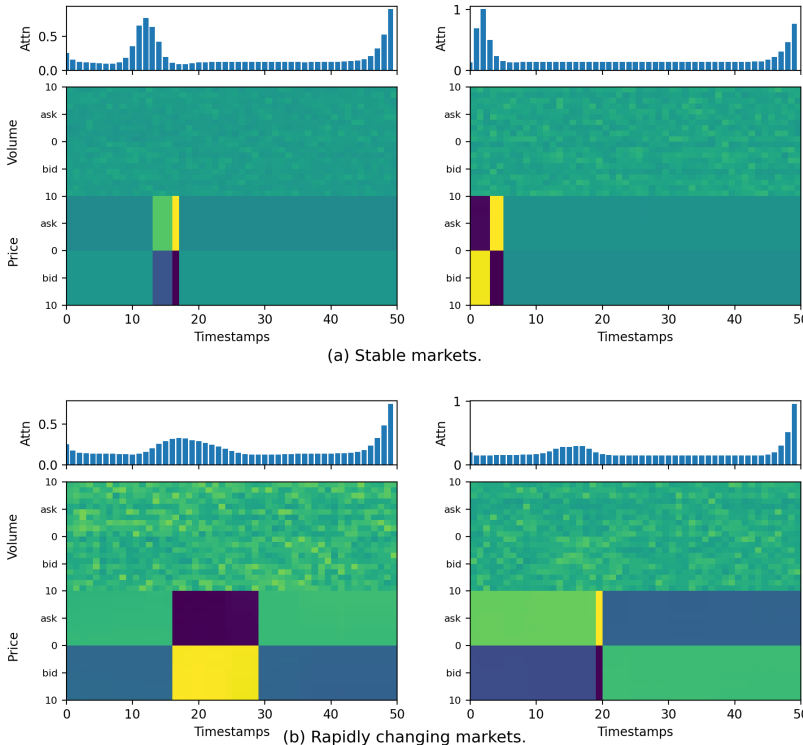


Figure 3: Attention visualization for SOFT-AS_{low}^{0.10} (stock 000858). (a) Stable markets: attention is concentrated on recent events. (b) Rapidly changing markets: attention spreads to earlier events.

E Exploratory history and additional tables

Before the matched 400k batch, three exploratory stages were run. (i) An initial 200k AS-guided comparison showed soft-AS nearly matching C-PPO while improving profit-ratio-like behaviour, BC warm start positive but weaker, and hard constraints too restrictive. (ii) A profit-only correction confirmed and removed a residual reward-shaping component that had leaked into an early soft-AS implementation: the regularizer must not smuggle the old shaping into the base objective. (iii) A 32-run sweep over λ , γ , κ , and hard-window widths indicated that teacher calibration matters more than small λ changes, motivating the final design.

The headline paper reports one selected soft variant, but the broader sweep was used to check sensitivity. With the base teacher, mean PnL for $\lambda = \{0.03, 0.05, 0.10, 0.125, 0.20\}$ was $\{64.7, 73.5, 69.3, 76.6, 68.0\}$ and mean absolute inventory was $\{43.5, 45.4, 54.5, 76.1, 68.5\}$. This supports a broad useful range rather than a single brittle value. Table 4 gives per-stock means for the reported methods.

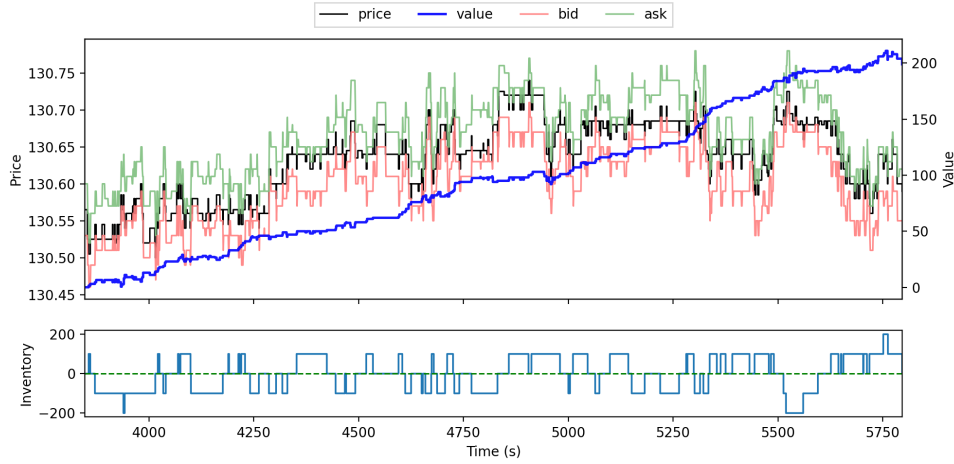


Figure 4: Decision trace of the proposed $\text{SOFT-AS}_{\text{low}}^{0.10}$ policy on one test episode. Top: mid-price (black), quoted bid/ask (red/green), cumulative value (blue). Bottom: inventory, bounded around zero. The policy quotes actively around a drifting fair value while controlling inventory.

Table 4: Per-stock mean test PnL for the reported methods. PPO methods use three seeds per stock; direct AS is deterministic and run once per stock.

Method	000001	000858	002415
C-PPO (reward-shaped)	0.74	169.62	8.67
PROFIT-PPO (control)	2.52	183.45	12.12
$\text{SOFT-AS}_{\text{low}}^{0.10}$ (proposed)	9.67	186.41	25.80
Hard-AS $w=0.30$	4.38	166.39	14.20
AS direct (fitted)	10.36	34.41	20.64